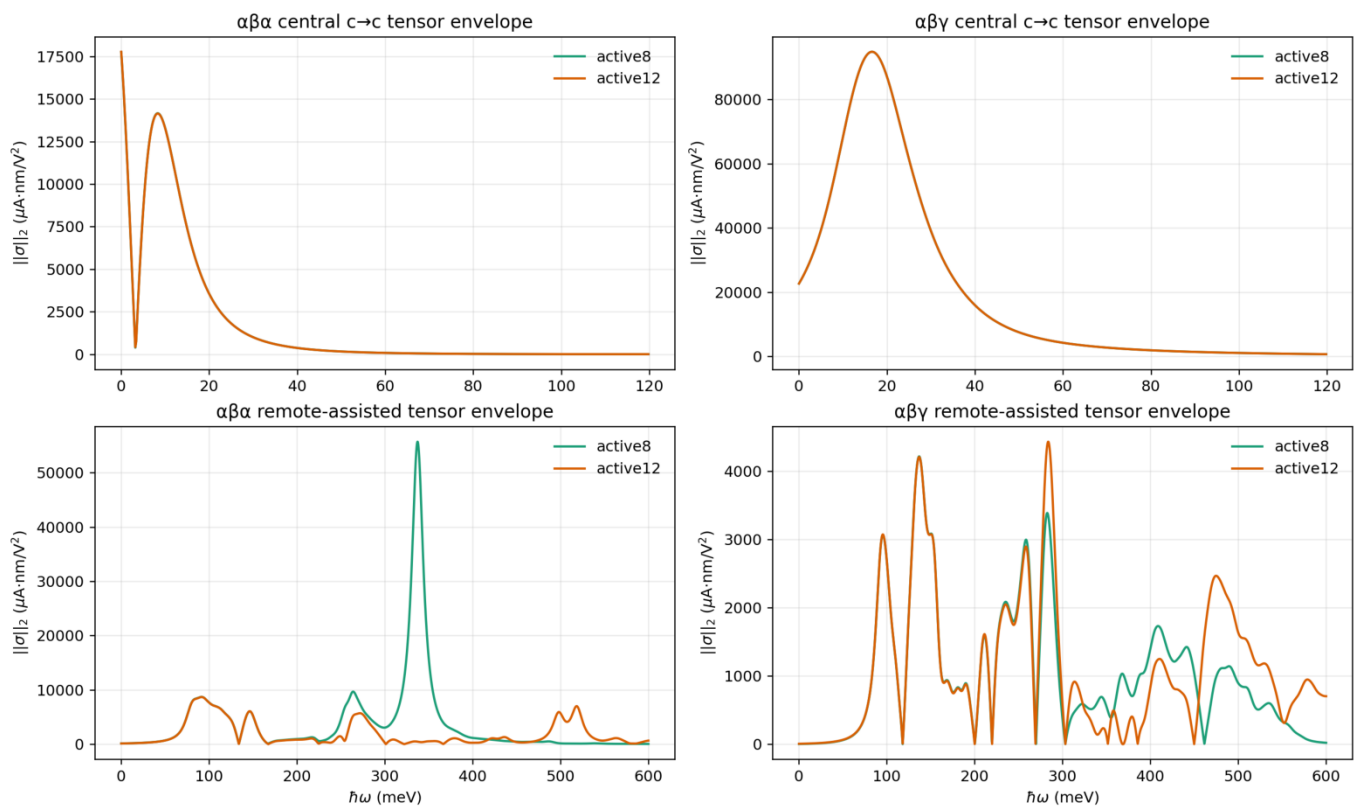


HTQG 巨大体光伏效应

拓扑、序参量与实验主叙事

Type-II bright moire vs Type-I control



- 核心事实：Type-II $\alpha\beta\gamma$ 的 clean finite-frequency BPVE 远大于 Type-I $\alpha\beta\alpha$ 。
 - $8\text{e}4\text{--}1\text{e}5$ historical scale 来自 17 meV 左右的 central c_{to}c resonance，
不是 crossing / near-degeneracy anomaly。
 - Chern 与 order 会重组 Bloch geometry，但不会量子化或单独决定 BPVE magnitude。
 - 实验主轴：domain-resolved、polarization-resolved THz photocurrent。
- 2026-08-04 | figure-led report | no raw data included

“巨大 BPVE”在本项目中的严格含义

Shift-current observable

$$j_{2D}^a(0) = 2 \sigma^{a;bc}(0; \omega, -\omega) \text{Re}[E_b(\omega)E_c(-\omega)]$$

- Giant 不是把 Lorentzian width 调小后得到的单点最大值。
- 必须同时看到：finite frequency、finite direct gap、large integrated weight、
mesh stability、active-window stability，以及相对 matched moire control 的显著层级差。
- Shift current 由 resonant phase space、optical matrix elements、
gauge-covariant generalized derivative/shift factor 和 signed cancellation 共同决定。

Figure-scope normalization lineage

Figures 1-10 signed spectra/integrands: $\sigma_{\text{current}} = -\sigma_{\text{old}} / 2$

Figures 1-10 norms/envelopes/absolute weights: $Q_{\text{current}} = Q_{\text{old}} / 2$

JDOS, gaps, Chern/order labels, Berry/metric diagnostics: unchanged

Figure 11: already corrected; do not convert again

Current $\eta=8$ references: envelope norm $4.7397\text{e}4$; projected component $+2.3699\text{e}4$;
P2-approved continuum component $+2.3678\text{e}4$ microA nm/V² at 16.6 meV.

The few-thousand active6 Type-I results are controls, not the giant-BPVE headline.

一眼可见的主结果：Type-II 明显更亮

- 右上：Type-II $\alpha\beta\gamma$ historical envelope norm=9.4795e4; current norm=4.7397e4。
- 左上：Type-I $\alpha\beta\alpha$ 的 central response 小得多，并与 gapless low-frequency structure 纠缠。
- $\alpha\beta\gamma$ active8/active12 曲线重合；Type-I 的大 remote peak 不通过同一 active-window test。

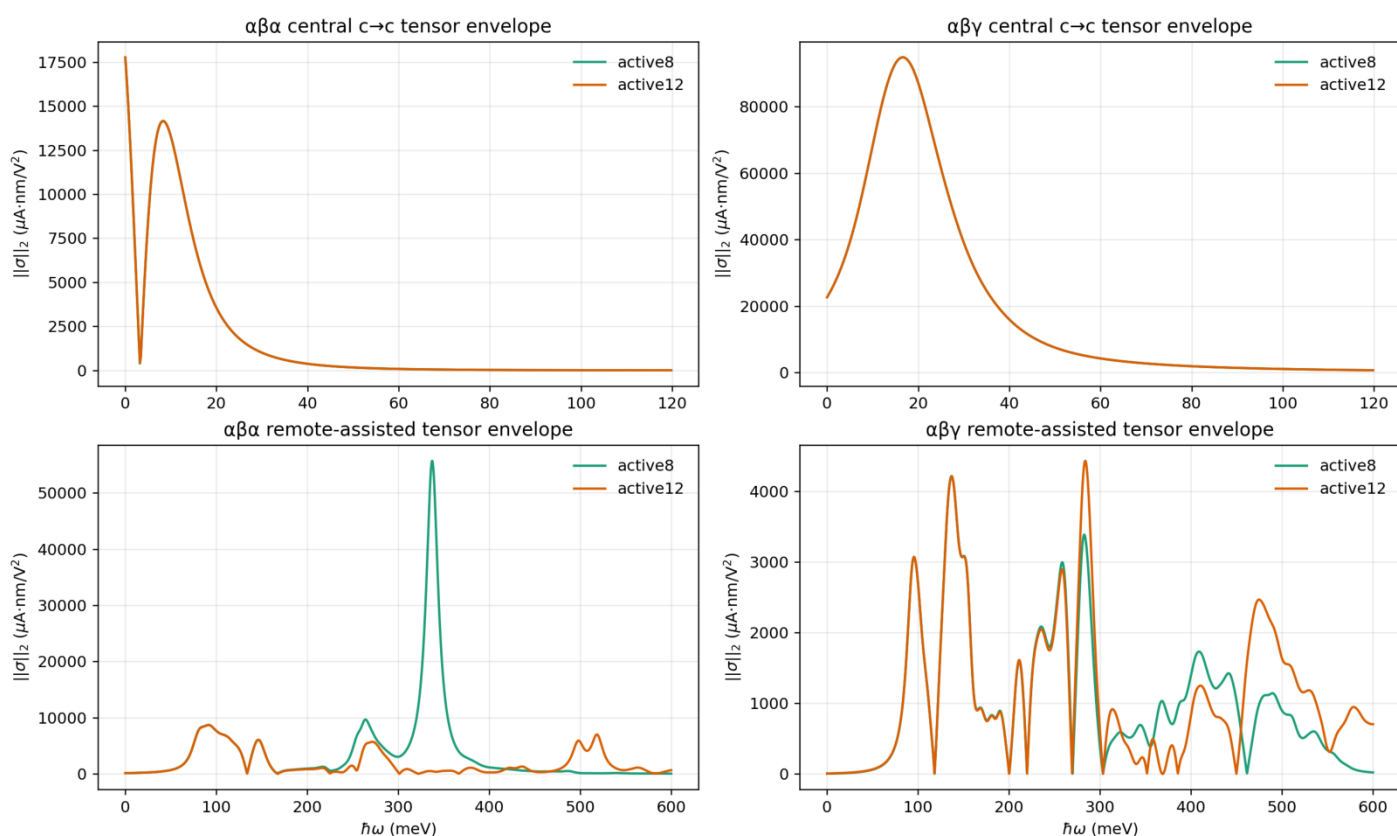


图 1 | 同一历史 convention 下的 moire-type contrast 与 active-window test。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

8e4-1e5 来自 finite-frequency resonance

- $\eta=2/3$ meV historical peaks 约 9.9e4/8.4e4; current magnitudes 约 4.97e4/4.19e4。
- 峰位始终约 16.6-17.8 meV; 不是 $\omega \rightarrow 0$ crossing spike。
- source direct/indirect gaps = 3.4795/1.5630 meV。
- P2 dominant source 是 singleton-to-singleton, candidate_near_degenerate_group=false。
- Arbitrary near-degenerate U(N) material response 仍在 approved scope 之外。

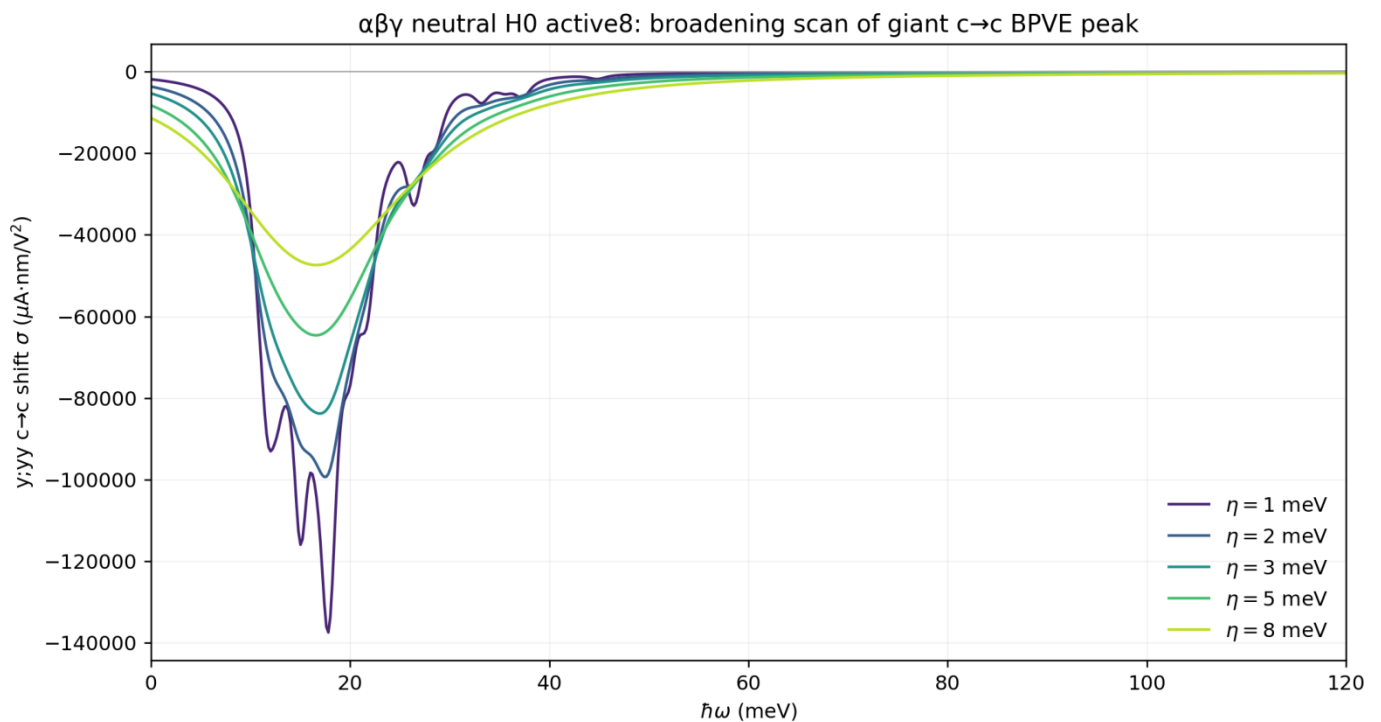


图 2 | $\alpha\beta\gamma$ neutral-H0 central c_to_c broadening scan。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

18×18、24×24、30×30 曲线重合

- 三套 k mesh 的 peak frequency、line shape 与 integrated weight 一致。
- 与 finite direct gap 和 active8→active12 overlap 一起，排除离散网格偶然热点解释。
- 这不是对所有 microscopic terms 的批准；它验证的是当前 projected H0 response path。

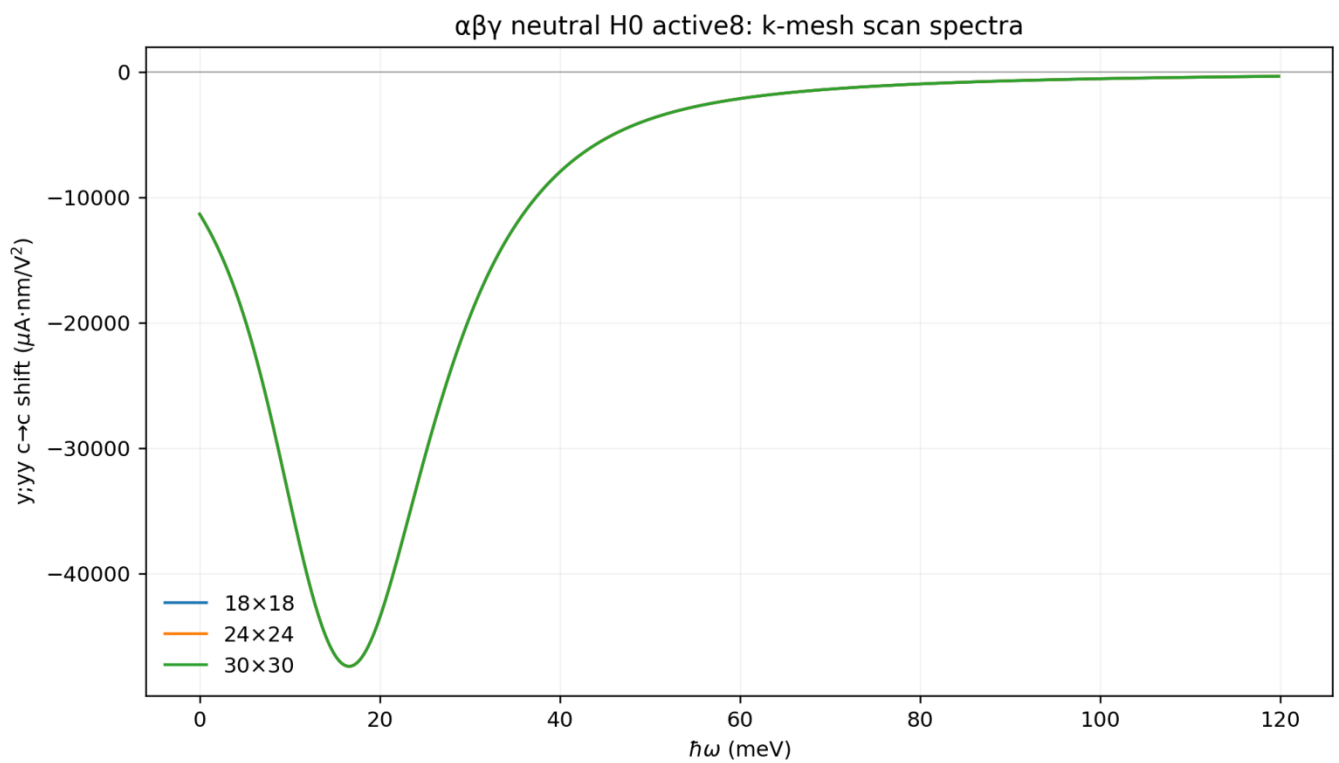


图 3 | $\alpha\beta\gamma$ $\eta=8$ meV k-mesh spectra.

Figures 1-10: signed response -> -old/2; norms/absolute weights -> old/2

巨大响应是 signed tensor，不是单一正数

- 多个 symmetry-related components 同时很大，并具有相反符号。
- 下图 central c_to_c 几乎重现低能 total：不是 remote-band artifact。
- 实验必须指定 polarization 与 current axis；tensor norm 只用于结构层级比较。

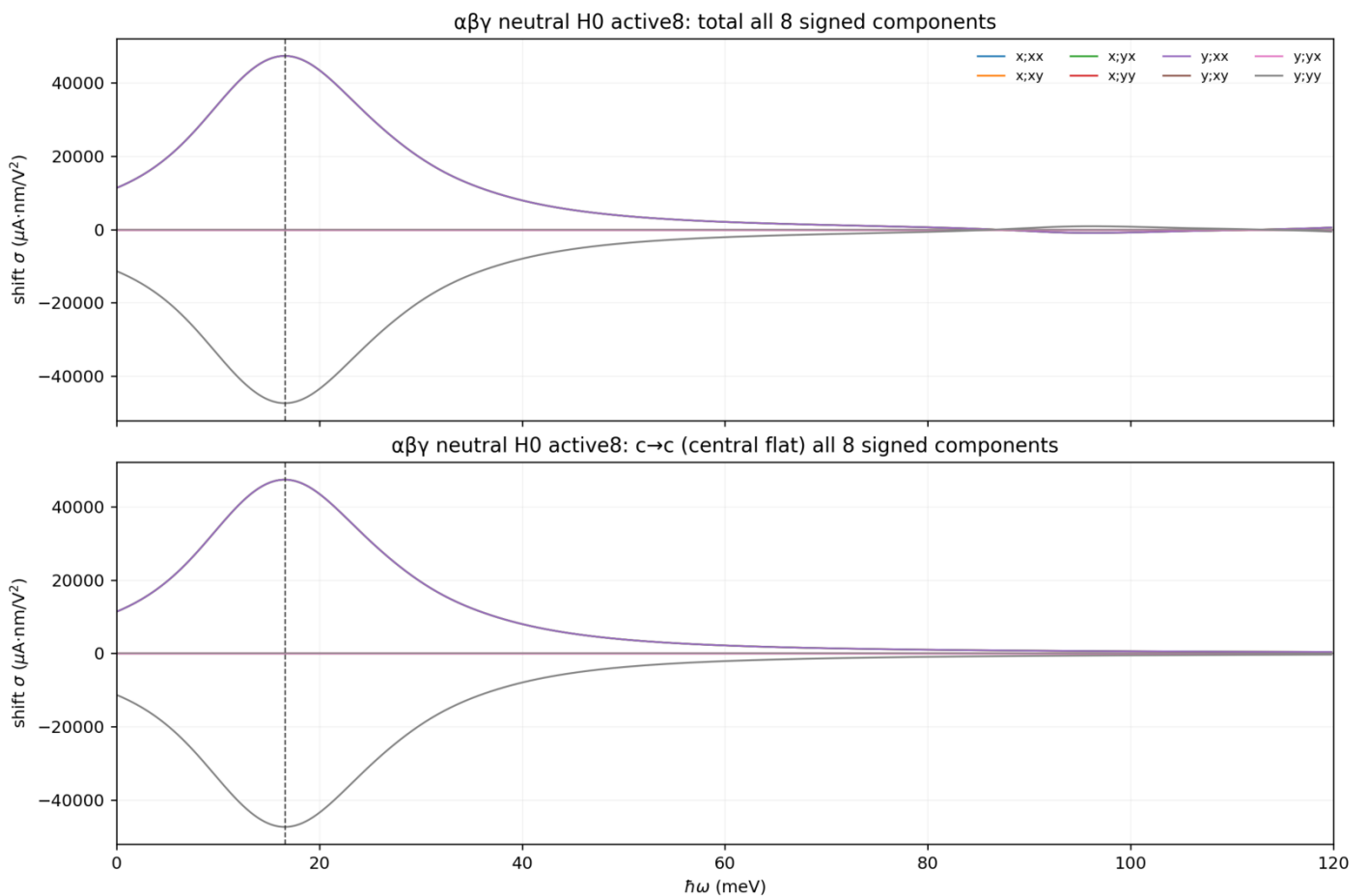


图 4 | Type-II αβγ 的全部八个 signed shift-current components。

Figures 1-10: signed response -> -old/2; norms/absolute weights -> old/2

Mechanism : central-flat resonance , 而非 remote artifact

- JDOS 低能峰由 $c \rightarrow c$ transitions 主导；dominant current component 也由同一 channel 重现。
- JDOS 只说明 resonant phase space；巨大 amplitude 还需要 optical weight、
shift/geometric factor 与有限 signed cancellation。
- 这是“哪组跃迁”答案，不是“哪个单一几何量造成 amplitude”答案。

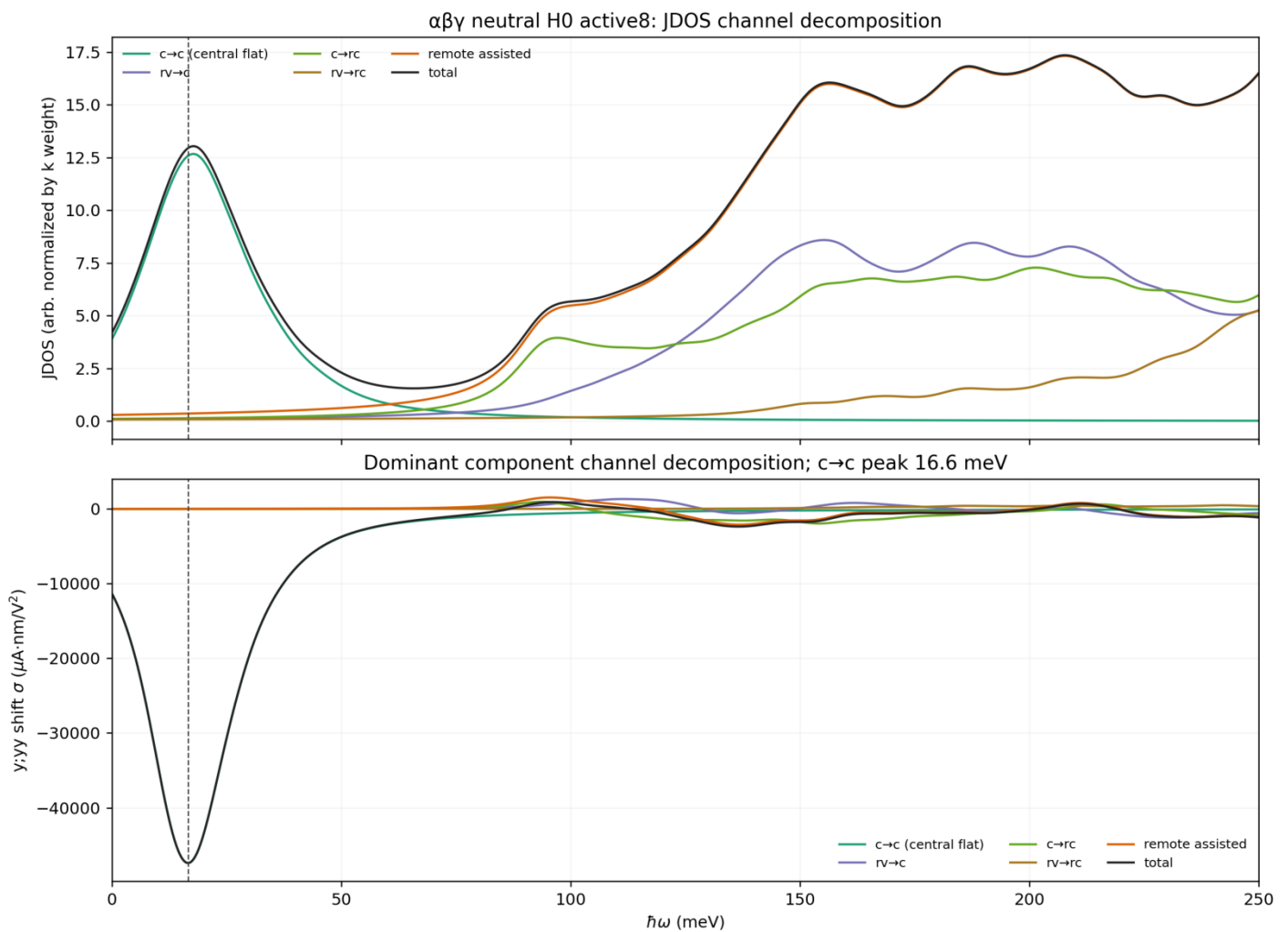
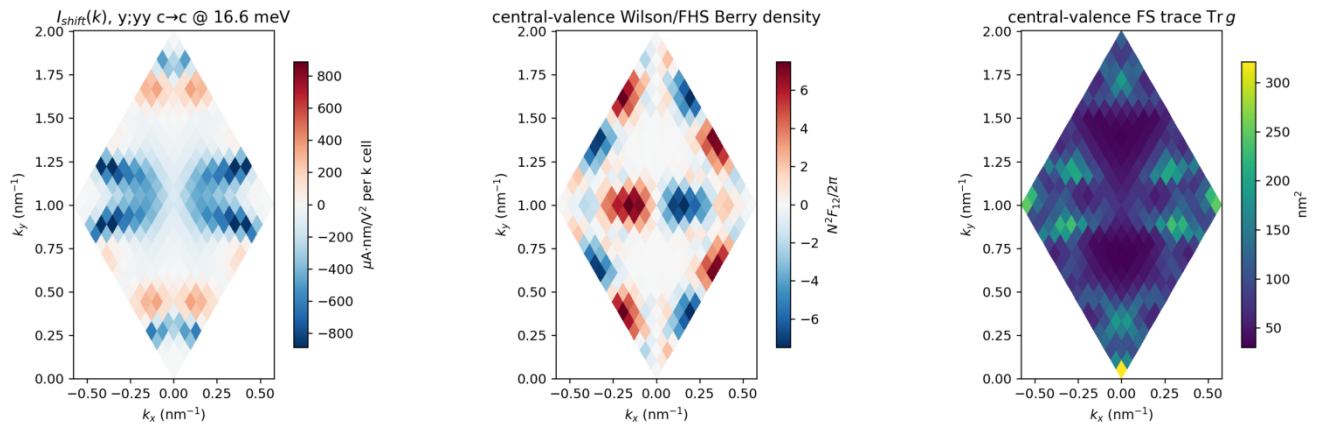


图 5 | 上: JDOS channels; 下: dominant component channels.

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

Local quantum geometry 是共同底层，不是 Chern 定标

- Shift-current、Berry-curvature density 与 FS trace 都在 mini-BZ 中呈现强烈结构。
- Berry/FS maps 是 occupied-bundle diagnostics，不是 pair-resolved transition geometry。
- FS trace 是 finite-difference subspace-fidelity diagnostic；co-location 不是 causal proof。



$\text{Chern} = (1/2\pi) \int_{\text{BZ}} \Omega(k) dk$ 是 unweighted global integral。

BPVE 则是 frequency-resolved、transition-weighted、tensorial signed integral。
 现有比较只证明 Chern alone 不决定 magnitude，不建立 universal necessary/sufficient criterion。

图 6 | 同一 $\alpha\text{-Bi}_2\text{O}_3$ source 的 shift hotspot、Berry density 与 FS trace。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

HF order 重塑频率、符号、polarization 与 channel

- Type-II HF states 保留 bright central c_to_c; Type-I positive fillings 更弱或转为 c_to_rc。
- Order 改变 gaps、occupations、selection rules、wavefunction geometry 与 hotspot cancellation。
- BPVE 可提供 order-sensitive candidate discriminants; 单个大峰不能唯一识别 order。

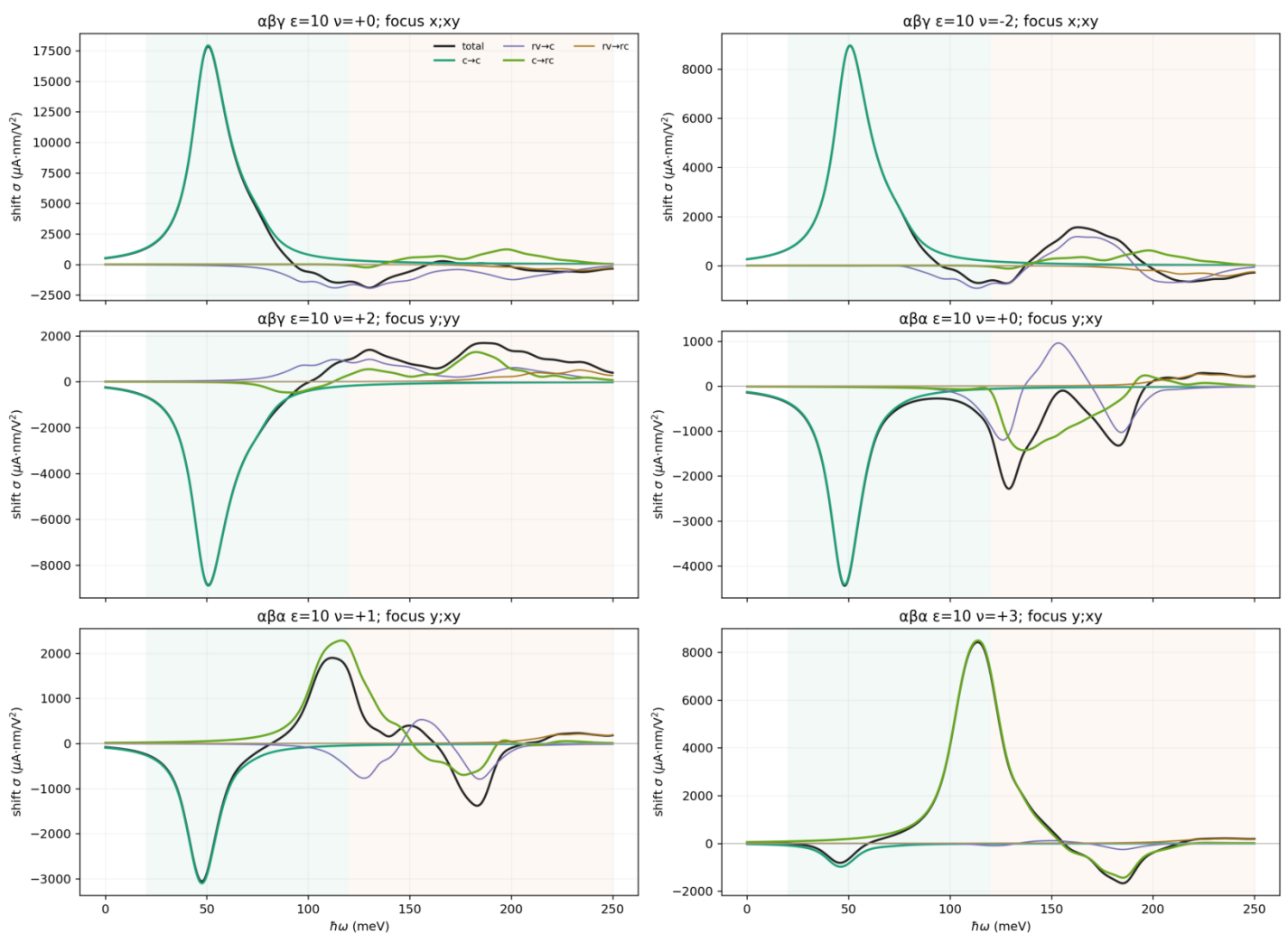


图 7 | active4 $\epsilon=10$ representative HF states 的 channel decomposition。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

Topology 与 order 相关，但 brightness 连续变化

- Type-II shown rows 均有 active2/active4 Chern agreement; 这是 active-space consistency.
- 它不是 thermodynamic phase stability; Type-I 集中了 Chern mismatches 与 IVC/KIVC order.
- $\epsilon=5$ active4 KIVC/TIVC 未计算; $\alpha\beta\alpha$ $v=0$ order amplitudes 有 inherited-table caveat.

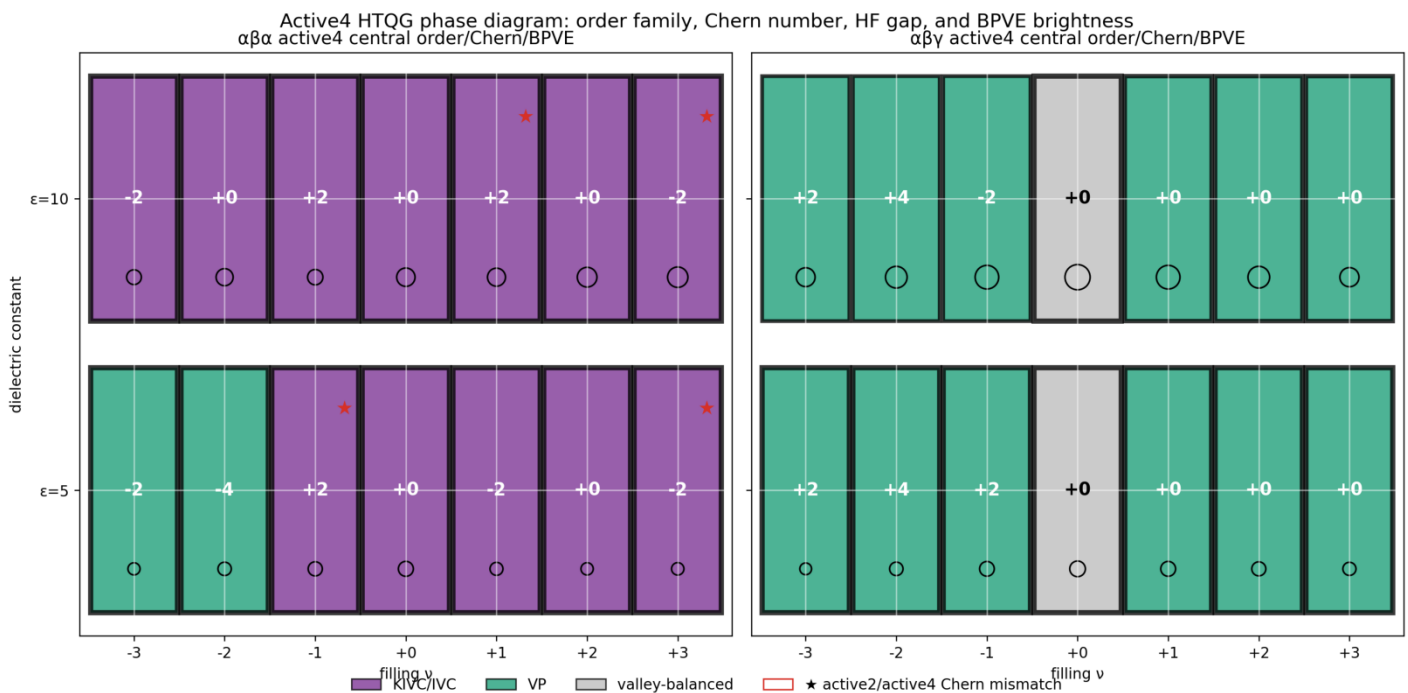


图 8 | Background=order, 数字=Chern, 边框=gap, 空心圆=low-energy BPVE weight.

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

固定 Chern/order 内，BPVE 仍能强烈变化

- 每条 sampled continuation branch 的 Chern sector 与 reported order family 保持不变。
- $\varepsilon=8/12/15$ seeded from $\varepsilon=10$; branch crossings 与 global ground state 未测试。
- BPVE 随 gap softening 强变; wavefunction-geometry change plausible, 但该 scan 未直接计算。

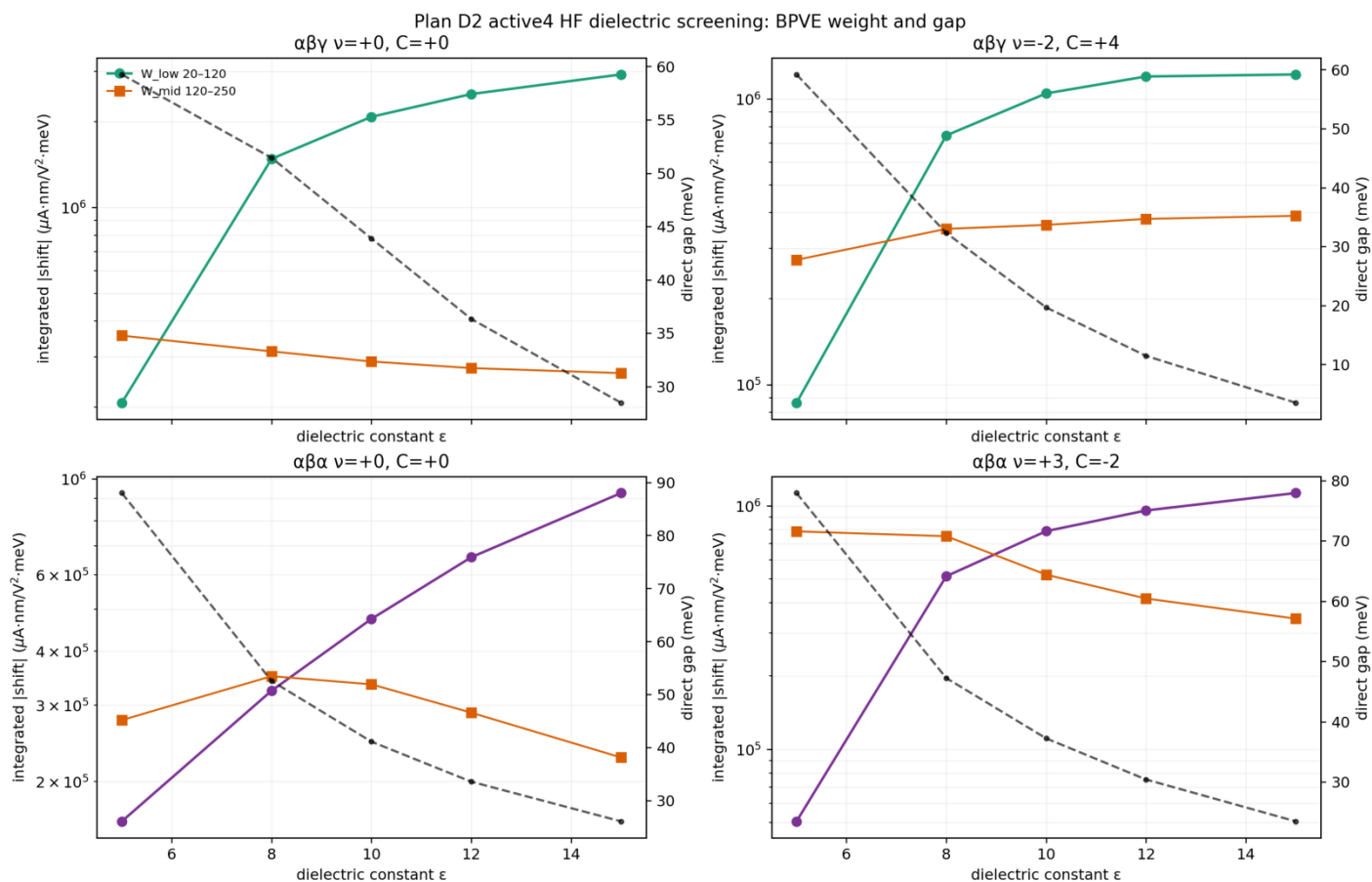


图 9 | 四条 active4 HF branches 的 dielectric scan。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

单畴巨大，等面积 partner domains 几乎抵消

- Type-II $\alpha\beta\gamma$ / $\gamma\beta\alpha$ 是近乎 equal-and-opposite rank-3 tensors。
- Equal-area lab-frame average 接近零，不代表每个 domain 不亮。
- 实验应使用 single-domain / imbalanced-domain sample 或 local photocurrent mapping。

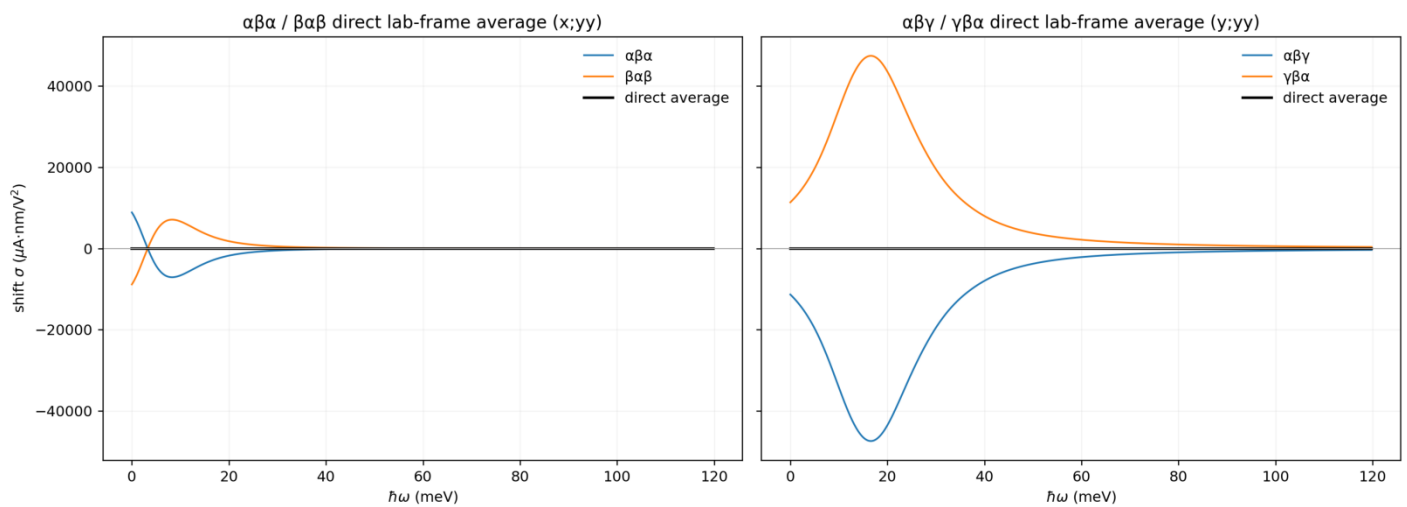


图 10 | Type-I 与 Type-II structural partner-domain direct averages。

Figures 1-10: signed response $\rightarrow$ -old/2; norms/absolute weights $\rightarrow$ old/2

实验主轴：polarization-resolved、domain-resolved THz current

- 16.6 meV 约对应 4 THz；HF order/screening 可把强峰移动到 tens-of-meV window。
- 测量 peak frequency、signed current、polarization phase 与 partner-domain reversal。
- Type-II row 是 clean insulating projected estimate；Type-I row 只作 gapless coefficient diagnostic。
- 图中 current conversion 已采用 reviewed -old/2 correction，但仍不是 measured-device prediction。

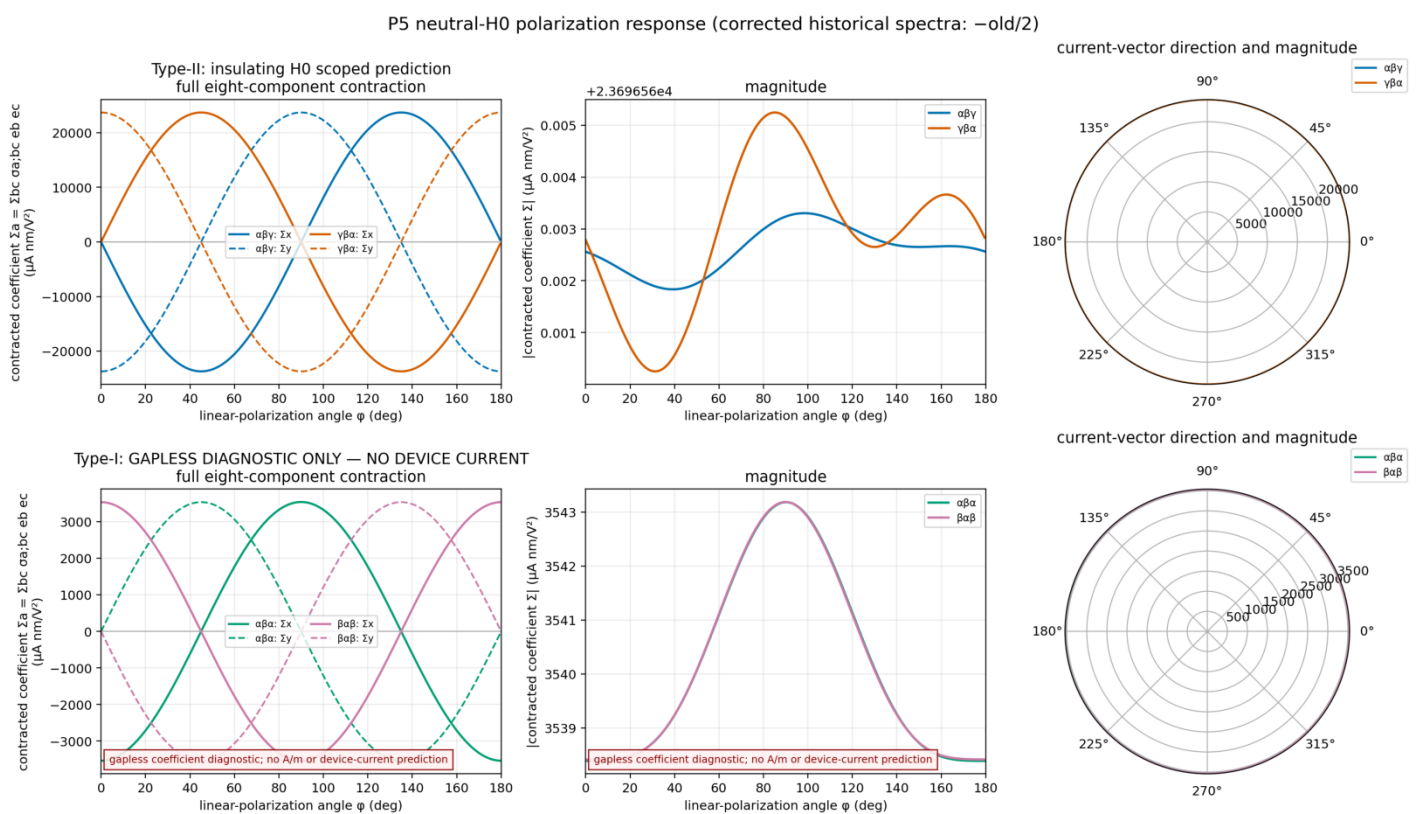


图 11 | Full-tensor polarization contraction 与 current-vector direction。

重写后的核心叙事

Six-step story

- Structural selectivity: Type-II $\alpha\beta\gamma$ 远亮于 Type-I $\alpha\beta\alpha$ 。
- Numerical legitimacy: finite gap/frequency + broadening + mesh + active-window tests。
- Mechanism: central c_to_c resonant weight $\times$ optical/geometric factor $\times$ signed cancellation。
- Topology: Chern 是 global label; BPVE 对 transition-weighted local geometry 敏感。
- Order: 可重塑 gap/sign/channel, 并为 independently validated branches 提供 candidate discriminants。
- Experiment: domain-resolved、polarization-resolved THz photocurrent 是首要 endpoint。

Immediate research program

- Matched Type-I/II factorization: JDOS、 $|r|^2$ 、covariant shift term、cancellation。
- Topology controls: same-order/different-C 与 same-C/different-order pairs。
- Order spectroscopy: multiseed HF branches, 比较 peak/sign/polarization/channel。
- Experimental scan: domain、gate filling、dielectric environment、temperature。

Active6 Type-I P1-P5 = controlled comparison / supplement, not giant-BPVE headline.

边界与不可跨越的表述

- Figures 1-10: signed response 用 $-\text{old}/2$; norm/absolute weight 用 $\text{old}/2$; Figure 11 不再转换。
- Neutral Type-I H0 low-frequency response 是 gapless diagnostic, 不是 clean insulating giant BPVE。
- Type-II 是 projected continuum-envelope internal-interband resonant result;
orbital-position/external terms 和 exact finite-width antiresonant tails 未包含。
- Berry/metric hotspot 共定位不等于单因子 causal proof。
- Domain fractions 不是 thermodynamic population; ideal current 不是 measured-device current。
- Chern alone 不决定 shift-current amplitude; 现有结果不建立 universal necessary/sufficient criterion。

Selected references

Sipe & Shkrebtii, Phys. Rev. B 61, 5337 (2000).

Morimoto & Nagaosa, Sci. Adv. 2, e1501524 (2016).

Chaudhary et al., Phys. Rev. Research 4, 013164 (2022).

Package content and scope

Figures 1-10: historical Plan-B products. Figure 11: corrected P5 product.

Only $\alpha\beta\gamma$ is directly P2 full-continuum approved; $\gamma\beta\alpha$ is projected-active8 partner.

Narrative + exact figures only. No NPZ/JSON/CSV/checkpoints/HF density.